

Information Coefficient (IC) Reference

Mathematical Foundations for Signal Quality Assessment

NAT Quantitative Research Platform

NAT Research

June 2026

Contents

1	Introduction	2
2	Core Definitions	2
2.1	Forward Log Return	2
2.2	Spearman Rank IC	2
2.3	Rolling Non-Overlapping IC	3
3	Derived Metrics	3
3.1	IC Mean and Standard Deviation	3
3.2	Information Ratio (IC IR)	3
3.3	Annualized IC IR	3
3.4	Statistical Significance	4
3.4.1	t-Statistic and p-Value (Window-Based)	4
3.4.2	Z-Test p-Value (Observation-Based)	4
3.5	IC Hit Rate	4
3.6	IC Autocorrelation	4
3.7	Trading Cost Metrics	4
3.7.1	Signal Turnover	4
3.7.2	Breakeven Fee (bps)	5
4	Computation Variants	5
4.1	Rolling IC (Non-Overlapping)	5
4.2	Expanding IC (Causal)	5
4.3	Conditional IC (Regime-Gated)	5
4.4	Bandpass-Filtered IC	5
4.5	Tick-Level vs. Bar-Resampled Paths	6

5	Statistical Quality Control	6
5.1	Benjamini–Hochberg FDR Correction	6
5.2	5-Gate Hypothesis Protocol	6
5.3	Adaptive IC Threshold	7
6	IC Decay Monitoring	7
7	Regime-Conditioned Feature Weights	8
8	Threshold Reference	8
8.1	IC Computation Parameters	8
8.2	Agent Gate Thresholds	8
8.3	Decay and FDR Parameters	9
9	Algorithm Development Guide	9
9.1	Computing IC for a New Signal	9
9.2	Gate Checklist	9
9.3	Interpreting IC Values	10

1 Introduction

The Information Coefficient (IC) is the canonical metric for measuring a signal’s predictive power over future asset returns. In the NAT platform, IC governs every stage of the research pipeline: discovery, hypothesis gating, walk-forward validation, regime conditioning, and live decay monitoring.

This document provides the closed-form definitions, derived metrics, statistical tests, and threshold configurations as implemented in the NAT codebase. It is intended as a self-contained reference for developing new IC-based algorithms targeting Hyperliquid perpetual futures (BTC, ETH, SOL) across tick-level to macro horizons.

Source files:

- `scripts/exploration/spannung_regime_screener.py` — core IC computation
- `scripts/alpha/screener.py` — rolling/expanding IC, FDR correction
- `scripts/agent/gates.py` — gate protocol (G1, G2)
- `scripts/agent/base.py` — decay monitoring
- `scripts/alpha/regime_filter.py` — conditional IC
- `config/agent.toml` — threshold configuration

2 Core Definitions

2.1 Forward Log Return

Definition 1 (Forward Return). *Given a price series $\{p_t\}$ and a horizon h (in ticks or bars), the forward log return at time t is:*

$$r_{t \rightarrow t+h} = \ln \left(\frac{p_{t+h}}{\max(p_t, \varepsilon)} \right) \quad (1)$$

where $\varepsilon = 10^{-15}$ prevents division by zero. The final h entries are set to NaN (no forward return beyond the array boundary).

Remark 1. *The horizon h is expressed in native sampling units. At tick level (100 ms emission), $h = 10$ corresponds to a 1 s horizon; $h = 50$ corresponds to 5 s. For bar-resampled data, h is in bars.*

2.2 Spearman Rank IC

Definition 2 (Information Coefficient). *The IC of a signal S at horizon h is the Spearman rank correlation between the signal value at time t and the forward return:*

$$\text{IC} = \rho_s(S_t, r_{t \rightarrow t+h}) = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)} \quad (2)$$

where $d_i = \text{rank}(S_i) - \text{rank}(r_i)$ and n is the number of valid (non-NaN) paired observations.

Remark 2. *Spearman rank correlation is preferred over Pearson because: (i) it is robust to outliers common in microstructure data, (ii) it captures monotonic (not just linear) relationships, (iii) it is invariant to monotone transformations of the signal. The sole exception in NAT is the Kalman filter tuning module, which uses Pearson correlation for filter parameter optimization.*

2.3 Rolling Non-Overlapping IC

In practice, IC is computed over rolling non-overlapping windows to ensure statistical independence between estimates.

Given N observations, partition into $K = \lfloor N/W \rfloor$ windows of size W . For window k :

$$\text{IC}_k = \rho_s\left(S_{kW:(k+1)W}, r_{kW:(k+1)W}\right) \quad (3)$$

A window is **valid** if and only if:

1. Number of finite paired observations $\geq n_{\min}$ (default: 100)
2. Signal range $\text{ptp}(S) > 10^{-15}$ (not constant)
3. Return range $\text{ptp}(r) > 10^{-15}$ (not constant)
4. Spearman computation does not return NaN

Default parameters: $W = 3000$ ticks, $n_{\min} = 100$.

3 Derived Metrics

Given K valid IC windows $\{\text{IC}_1, \dots, \text{IC}_K\}$:

3.1 IC Mean and Standard Deviation

$$\overline{\text{IC}} = \frac{1}{K} \sum_{k=1}^K \text{IC}_k \quad (4)$$

$$\sigma_{\text{IC}} = \sqrt{\frac{1}{K} \sum_{k=1}^K (\text{IC}_k - \overline{\text{IC}})^2} \quad (5)$$

3.2 Information Ratio (IC IR)

Definition 3 (Information Ratio). *The IC IR measures signal-to-noise of predictability:*

$$\text{IR} = \frac{\overline{\text{IC}}}{\sigma_{\text{IC}}} \quad (6)$$

A signal with $\text{IR} > 1$ has mean IC exceeding its variability — a strong indicator of persistent predictive power.

3.3 Annualized IC IR

To compare signals across timeframes, the IR is annualized:

$$\text{IR}_{\text{ann}} = \text{IR} \times \sqrt{\frac{B_{\text{year}}}{B_{\text{step}}}} \quad (7)$$

where B_{year} is the number of bars per year and $B_{\text{step}} = N_{\text{bars}}/K$ is the average bars per IC window.

Timeframe	B_{year}	Computation
15 min	35,040	96×365
1 h	8,760	24×365
4 h	2,190	6×365

3.4 Statistical Significance

3.4.1 t-Statistic and p-Value (Window-Based)

For the null hypothesis $H_0: \overline{\text{IC}} = 0$:

$$t = \frac{\overline{\text{IC}}}{\sigma_{\text{IC}}/\sqrt{K}} \quad (8)$$

$$p = 2 \Pr(|T| > |t|), \quad T \sim t_{K-1} \quad (9)$$

where t_{K-1} is the Student's t -distribution with $K - 1$ degrees of freedom.

3.4.2 Z-Test p-Value (Observation-Based)

For a single IC estimate over n observations, under $H_0: \text{IC} = 0$:

$$p = \text{erfc}\left(\frac{|\text{IC}| \cdot \sqrt{n}}{\sqrt{2}}\right) \quad (10)$$

where erfc is the complementary error function. This approximation treats the Fisher-transformed Spearman correlation as asymptotically normal.

3.5 IC Hit Rate

$$\text{HitRate} = \frac{1}{K} \sum_{k=1}^K \mathbf{1}[\text{sign}(\text{IC}_k) = \text{sign}(\overline{\text{IC}})] \quad (11)$$

A hit rate above 70% indicates the signal consistently predicts in the same direction across windows.

3.6 IC Autocorrelation

Lag-1 autocorrelation of the IC series:

$$\rho_1 = \frac{\sum_{k=1}^{K-1} (\text{IC}_k - \overline{\text{IC}})(\text{IC}_{k+1} - \overline{\text{IC}})}{\sum_{k=1}^K (\text{IC}_k - \overline{\text{IC}})^2} \quad (12)$$

Low autocorrelation ($|\rho_1| < 0.3$) suggests IC is noisy window-to-window; high autocorrelation suggests persistent regimes of predictability.

3.7 Trading Cost Metrics

3.7.1 Signal Turnover

$$\text{Turnover} = \frac{\text{mean}(|\Delta S_t|)}{\sigma(S)} \quad (13)$$

where $\Delta S_t = S_t - S_{t-1}$. High turnover implies frequent rebalancing and higher transaction costs.

3.7.2 Breakeven Fee (bps)

The minimum fee level at which the signal remains profitable:

$$\text{Breakeven}_{\text{bps}} = \frac{|\overline{\text{IC}}| \times \sigma(r)}{\text{Turnover}} \times 10,000 \quad (14)$$

Remark 3. *On Hyperliquid, taker fees are ~ 3.5 bps and maker fees are ~ 0.2 bps. A signal is cost-viable if $\text{Breakeven}_{\text{bps}}$ exceeds the relevant fee tier. When $\text{Turnover} \approx 0$, the breakeven is $+\infty$ (the signal never trades).*

4 Computation Variants

4.1 Rolling IC (Non-Overlapping)

The standard method (Section 2.3). Windows are non-overlapping to ensure independence of IC estimates. Used in the spanning regime screener and alpha screener.

Parameters: window size W , minimum observations $n_{\min}$.

4.2 Expanding IC (Causal)

To avoid lookahead bias in walk-forward evaluation, IC is computed on an expanding window anchored at $t = 0$:

$$\text{IC}_j^{\text{exp}} = \rho_s(S_{0:t_j}, r_{0:t_j}), \quad t_j = n_{\min} + j \cdot \Delta t \quad (15)$$

where $\Delta t = \max(\lfloor N/20 \rfloor, 1)$ is the step size and $n_{\min} = 50$ is the minimum data before the first IC estimate.

This produces a monotonically growing sample size, simulating what a researcher would observe in real time.

4.3 Conditional IC (Regime-Gated)

For a binary regime mask $M_t \in \{0, 1\}$, the conditional IC is:

$$\text{IC}_{\text{cond}} = \rho_s(S_t \cdot M_t, r_{t \rightarrow t+h} \cdot M_t) \quad (16)$$

computed only on observations where $M_t = 1$. The window size adapts:

$$W_{\text{cond}} = \min\left(W, \max\left(\left\lfloor \frac{n_{\text{valid}}}{4} \right\rfloor, n_{\min}\right)\right) \quad (17)$$

If fewer than 2 valid windows exist, a single global Spearman IC is computed as a fallback.

Minimum regime observations: $n_{\text{regime}} \geq 500$.

4.4 Bandpass-Filtered IC

Signals are optionally filtered through an ultra-low bandpass before IC computation:

$$\tilde{S}_t = \text{BPF}(S_t; f_{\text{lo}}, f_{\text{hi}}) \quad (18)$$

Default band: $f_{\text{lo}} = 0.005 \text{ Hz}$, $f_{\text{hi}} = 0.1 \text{ Hz}$. At 10 Hz sampling (100 ms ticks), this extracts the slow-moving component with periods between 10 s and 200 s.

Filtered IC often exceeds raw IC by removing high-frequency noise while preserving the predictive component.

4.5 Tick-Level vs. Bar-Resampled Paths

IC computation supports two data representations:

Path	Forward Return	Horizon
Tick-level	$r_t = (p_{t+h} - p_t)/p_t$	$h = \max(1, \lfloor h_s/0.1 \rfloor)$ ticks
Bar-resampled	$r_t = (p_{t+h} - p_t)/p_t$	$h = \max(1, \lfloor h_s/T_{\text{bar}} \rfloor)$ bars

where h_s is the horizon in seconds and T_{bar} is the bar period in seconds.

5 Statistical Quality Control

5.1 Benjamini–Hochberg FDR Correction

When testing m hypotheses simultaneously, false discovery rate (FDR) control prevents spurious signals from surviving.

Algorithm (Benjamini–Hochberg, 1995):

1. Sort raw p-values: $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(m)}$

2. Compute adjusted p-values:

$$p_{(i)}^{\text{adj}} = p_{(i)} \cdot \frac{m}{i} \quad (19)$$

3. Enforce monotonicity (step-down):

$$p_{(i)}^{\text{adj}} \leftarrow \min(p_{(i)}^{\text{adj}}, p_{(i+1)}^{\text{adj}}), \quad i = m-1, \dots, 1 \quad (20)$$

4. Clip to $[0, 1]$

5. Reject H_i where $p_i^{\text{adj}} < q$

Default FDR threshold: $q = 0.05$.

Remark 4. *In the agent protocol, FDR is applied at the end of each research cycle across all hypotheses that passed individual gates. This controls the expected fraction of false discoveries in the promoted signal registry.*

5.2 5-Gate Hypothesis Protocol

Every candidate signal must pass a sequence of quality gates before promotion. The IC-related gates are:

Gate 1 — IC Gate:

$$\text{PASS} \iff |\text{IC}| \geq \text{IC}_{\min} \quad (21)$$

The p-value is computed via (10) for informational reporting but does not gate independently.

Gate 2 — ΔIC Gate:

$$\Delta\text{IC} = \text{IC}_{\text{gated}} - \text{IC}_{\text{baseline}}, \quad \text{PASS} \iff \Delta\text{IC} \geq \Delta\text{IC}_{\min} \quad (22)$$

This ensures regime conditioning adds genuine value over the ungated signal. Skipped when no regime gate is specified.

Gate 5 — Regime Improvement:

$$\text{PASS} \iff \exists r \text{ s.t. } \frac{|\text{IC}_r|}{|\text{IC}_{\text{global}}|} > 1.5 \quad (23)$$

At least one regime must improve IC by 50% over the global (all-data) baseline.

Full gate sequence:

Gate	Name	Metric	Threshold
G1	IC	$ \text{IC} $	$\geq \text{IC}_{\min}$
G2	ΔIC	$\text{IC}_{\text{gated}} - \text{IC}_{\text{base}}$	$\geq \Delta\text{IC}_{\min}$
G3	Cost	avg return per trade (bps)	≥ 0.1 bps
G4	Coverage	regime observation fraction	≥ 0.10
G5	Regime	$\max(\text{IC}_r / \text{IC}_g)$	> 1.5

After all gates, surviving hypotheses undergo BH-FDR correction (Section 5.1) at $q = 0.05$.

5.3 Adaptive IC Threshold

As the signal registry matures, the acceptance bar tightens:

$$\text{IC}_{\min}^{\text{adapt}} = \max\left(\text{IC}_{\text{floor}}, 0.8 \times \text{median}\left(\{\text{IC}_{\text{exp}}^{(i)}\}_{i \in \text{registry}}\right)\right) \quad (24)$$

This prevents the registry from accumulating marginal signals once strong ones have been found.

6 IC Decay Monitoring

Live signals are monitored for IC degradation. The decay protocol:

1. Compute rolling IC on the latest 2 days of data
2. Compare against expected IC at registration:

$$\text{IC}_{\text{threshold}} = \text{IC}_{\text{expected}} \times \delta \quad (25)$$

where $\delta = 0.5$ (decay ratio).

3. Track consecutive days below threshold:

$$D_t = \begin{cases} D_{t-1} + 1 & \text{if } \text{IC}_{\text{rolling}} < \text{IC}_{\text{threshold}} \\ 0 & \text{otherwise (recovery)} \end{cases} \quad (26)$$

4. Retire signal when $D_t \geq D_{\max}$ (default: 14 days)

Rolling IC computation:

- Tick-level path: horizon in 100 ms ticks, $h = \max(1, \lfloor h_s/0.1 \rfloor)$
- Bar-resampled path: aggregate to bars first (5 min or 1 h), then compute Spearman IC
- Minimum valid data points: 500
- IC history: rolling buffer of 30 daily entries

7 Regime-Conditioned Feature Weights

When a regime improves IC, feature weights are adjusted per-regime:

$$w_f^{(r)} = \begin{cases} \frac{|\text{IC}_f^{(r)}|}{\sum_{f'} |\text{IC}_{f'}^{(r)}|} & \text{if } \frac{|\text{IC}_f^{(r)}|}{|\text{IC}_f^{(\text{global})}|} > 1.5 \\ \frac{|\text{IC}_f^{(\text{global})}|}{\sum_{f'} |\text{IC}_{f'}^{(\text{global})}|} & \text{otherwise} \end{cases} \quad (27)$$

This allows the system to exploit regime-specific predictability while falling back to global weights when conditioning does not help.

8 Threshold Reference

8.1 IC Computation Parameters

Parameter	Context	Value
IC_WINDOW	Rolling window size (ticks)	3,000
IC_MIN_OBS	Min valid obs per window	100
MIN_REGIME_OBS	Min obs per regime condition	500
ULTRALOW_BAND	Bandpass filter (Hz)	(0.005, 0.1)
FWD_HORIZONS	Forward return horizons	1 s = 10t, 5 s = 50t
Expanding min_obs	First expanding IC point	50
Expanding step	Step size	$\max(\lfloor N/20 \rfloor, 1)$
Rolling min_obs (bars)	Alpha screener	30

8.2 Agent Gate Thresholds

Parameter	Micro	MF (5 min)	Macro (1 h)
$\text{IC}_{\min}$	0.10	0.08	0.07
$\Delta \text{IC}_{\min}$	0.05	0.03	0.02
Min coverage	0.10	—	—
Walk-forward sign consistency	0.80	—	—

Rationale: Lower thresholds at longer horizons compensate for fewer observations per day (~ 24 bars/day at 1 h vs. $\sim 36,000$ ticks/hour at tick level).

8.3 Decay and FDR Parameters

Parameter	Description	Value
δ (decay ratio)	Retire if rolling IC $< \delta \times$ expected	0.5
$D_{\max}$	Consecutive days before retirement	14
q (FDR)	Benjamini–Hochberg threshold	0.05
OOS min dates	Minimum out-of-sample periods	2
Min symbols	Cross-asset replication requirement	2

9 Algorithm Development Guide

9.1 Computing IC for a New Signal

To evaluate a candidate signal S :

1. **Compute forward returns** at target horizon using (1).
2. **Rolling IC**: partition into non-overlapping windows of size W and compute Spearman IC per window via (3).
3. **Aggregate**: compute $\overline{\text{IC}}$, σ_{IC} , IR, t-stat, p-value.
4. **Cost check**: compute turnover (13) and breakeven (14). Reject if breakeven < 3.5 bps (taker) or < 0.2 bps (maker).
5. **FDR**: if testing multiple signals, apply BH correction at $q = 0.05$ before declaring significance.

9.2 Gate Checklist

Before promoting a signal to the registry, verify:

- ☐ G1: $|\text{IC}| \geq \text{IC}_{\min}$ for the target agent tier
- ☐ G2: $\Delta \text{IC} \geq \Delta \text{IC}_{\min}$ (if regime-gated)
- ☐ G3: Average return per trade ≥ 0.1 bps after costs
- ☐ G4: Regime coverage $\geq 10\%$ of observations
- ☐ G5: At least one regime with IC ratio $> 1.5 \times$ global
- ☐ FDR: Survives BH correction across the hypothesis batch
- ☐ Cross-symbol: IC replicates on ≥ 2 symbols
- ☐ Walk-forward: $\geq 80\%$ sign consistency across OOS periods

9.3 Interpreting IC Values

IC	Interpretation
< 0.02	Noise — no signal
0.02–0.05	Weak — may survive in ensemble only
0.05–0.10	Moderate — viable with low costs
0.10–0.20	Strong — standalone signal candidate
> 0.20	Exceptional — verify not overfit

Remark 5. *IC values above 0.20 at tick level should be treated with suspicion. Common causes: lookahead bias, stale feature lagging price, or insufficient data. Always verify with expanding IC and cross-symbol replication before trusting high IC values.*